# Capture the Flag 2: Widgets

**Goal:** Turn the single-card application into a simple multi-section interface.

### Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

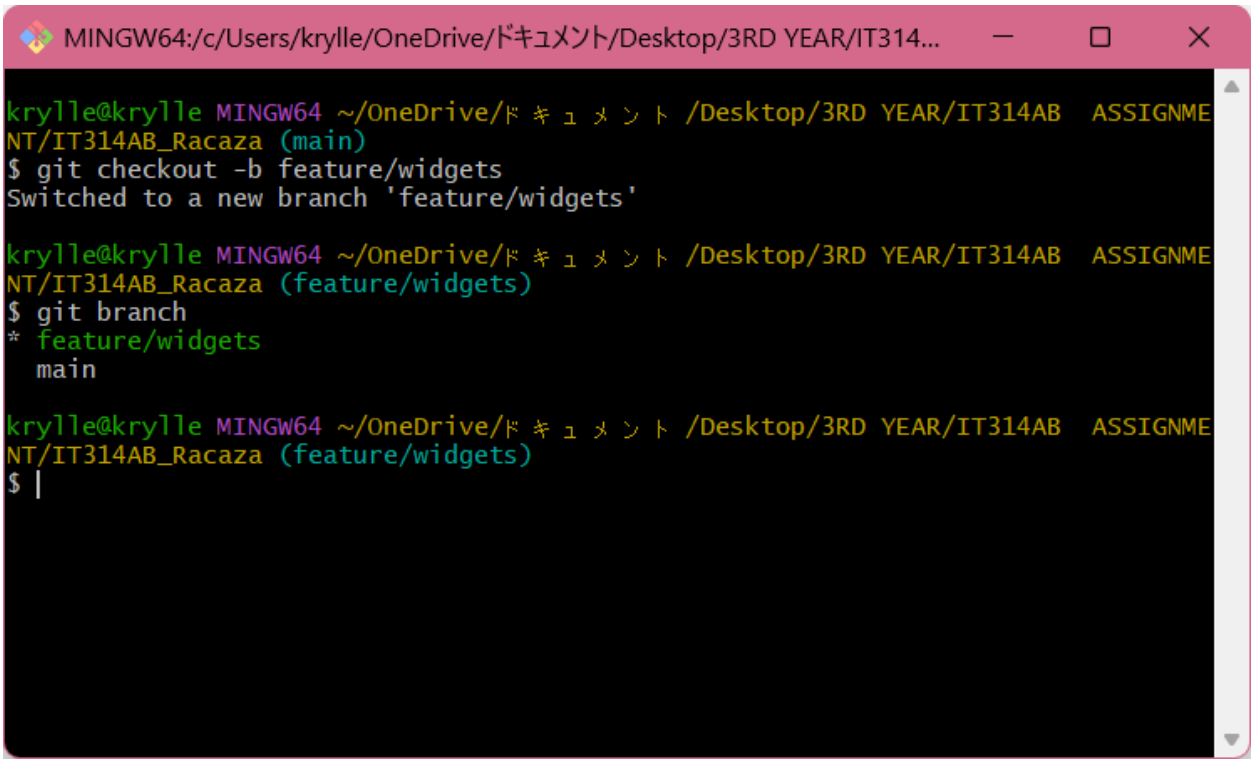
(I know what AI-generated code looks like, don't even think about it :>)

# 🚩 Flag 0 – Create Your Feature Branch

**Objective:** Create a feature branch before modifying your application. Name the branch feature/widgets

**Screenshot:**

- Terminal showing `feature/widgets`



# 🚩 Flag 1 – Break It Apart

**Objective:** Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

Text(

'Chiara Marie C. Canque\n'

'BSIT - 3\n'

'My First Flutter Application\n'

'August 11, 2026',

)

**Your task:** Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

Text → Date

**Do not worry about making it pretty yet.**

**Screenshot:**

- Your code showing the separate `Text` widgets
- Your application

```dart
    const Text(
      'Krylle Jane Y. Racaza',
      style: TextStyle(
        fontSize: 25,
        fontWeight: FontWeight.bold,
      ), // TextStyle
    ), // Text

    const SizedBox(height: 10),

    const Text(
      'BSIT - 3',
      style: TextStyle(fontSize: 20),
    ), // Text

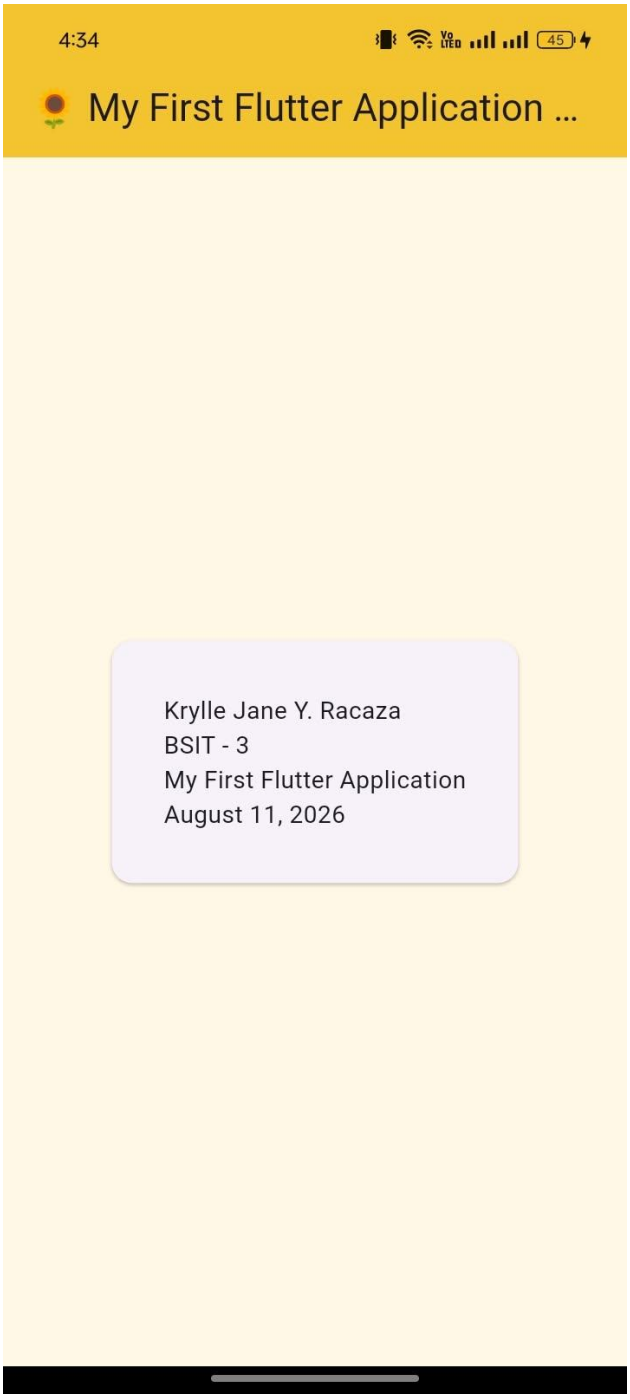
    const SizedBox(height: 10),

    const Text(
      'My First Flutter Application',
      style: TextStyle(fontSize: 18),
      textAlign: TextAlign.center,
    ), // Text

    const SizedBox(height: 10),

    const Text(
      'August 11, 2026',
      style: TextStyle(fontSize: 18),
    ), // Text

    const SizedBox(height: 10),
  ],
```



# 🚩 Flag 2 – Stack the Information

**Objective:** Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

Course & Section

Application Title

Date

**Screenshot:**

- The `Column` in your code
- The four separate pieces of information displayed vertically

```dart
            child: Column(
              mainAxisSize: MainAxisSize.min,
              children: [

                ClipOval(
                  child: Image.asset(
                    'assets/sunflower.png',
                    width: 130,
                    height: 130,
                    fit: BoxFit.cover,
                  ), // Image.asset
                ), // ClipOval

                const SizedBox(height: 15),

                const Text(
                  'Krylle Jane Y. Racaza',
                  style: TextStyle(
                    fontSize: 25,
                    fontWeight: FontWeight.bold,
                  ), // TextStyle
                ), // Text

                const SizedBox(height: 10),

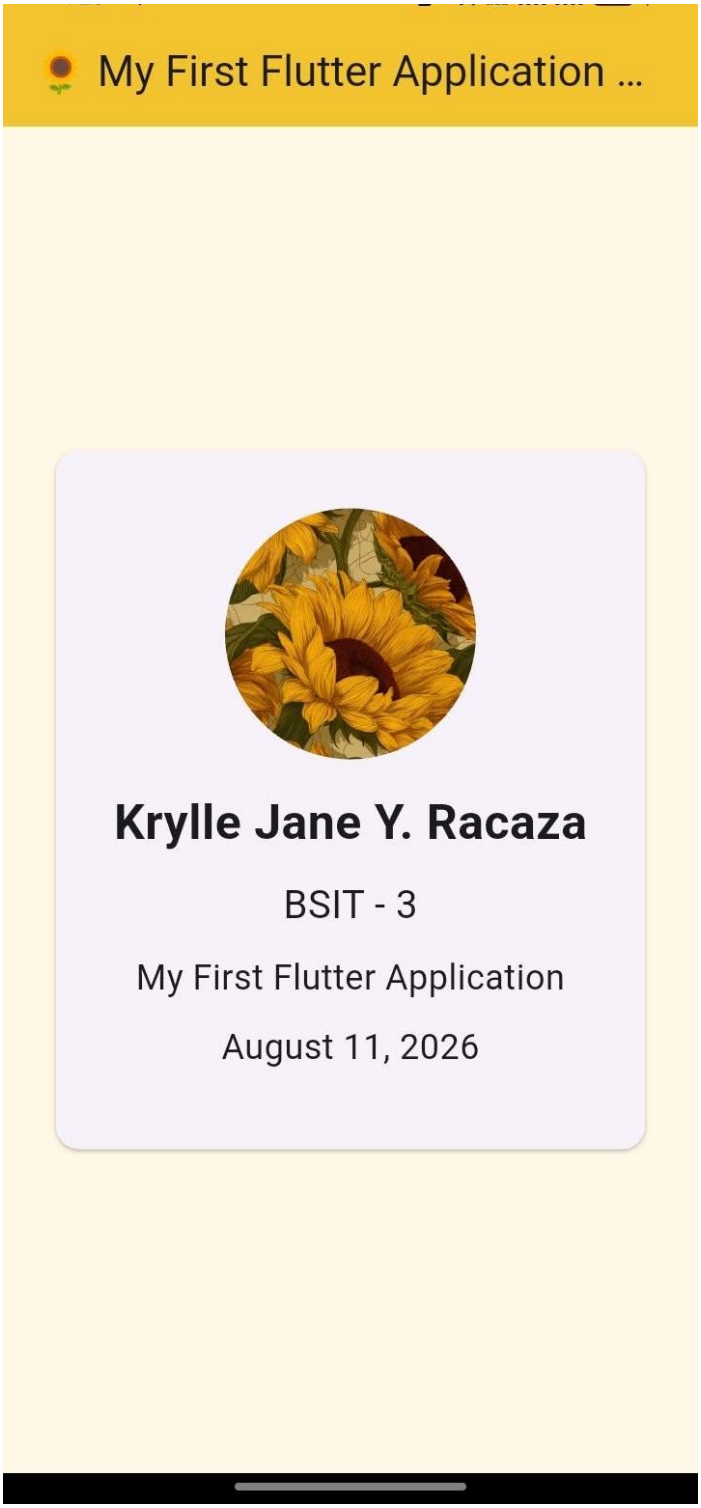
                const Text(
                  'BSIT - 3',
                  style: TextStyle(fontSize: 20),
                ), // Text

                const SizedBox(height: 10),
                const Text(
                  'My First Flutter Application',
                  style: TextStyle(fontSize: 18),
                  textAlign: TextAlign.center,
                ), // Text

                const SizedBox(height: 10),

                const Text(
                  'August 11, 2026',
                  style: TextStyle(fontSize: 18),
                ), // Text

                const SizedBox(height: 10),
                ),
              ],
            ), // Column
          ), // Padding
        ), // Card
      ), // Center
    ), // Scaffold
  ); // MaterialApp
}
```



# 🚩 Flag 3 – Add Another Element

**Objective:** Add another individual widget to your `Column`.

Add something that belongs to your profile.

Examples:

Hobby

It must be its **own widget**, not additional text inside another `Text`.

**Screenshot:**

- New widget in your `Column`
- Updated application



My First Flutter Application ...



**Krylle Jane Y. Racaza**

BSIT - 3

My First Flutter Application

August 11, 2026

Beauty and the Beast

# 🚩 Flag 4 – Put Two Things Side-by-Side

**Objective:** Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

Column

├── Text

├── Text

├── Text

├── Text

├── Text

└── Row

  ├── Text

  └── Text

**Screenshot:**

- Your widget structure/code
- The two items displayed side-by-side

```dart
Row(mainAxisAlignment: MainAxisAlignment.center,
  children: const [

    Text(
      'Beauty and the Beast',
      style: TextStyle(fontSize: 18),
    ), // Text

    SizedBox(width: 20),

    Text(
      'Belle',
      style: TextStyle(fontSize: 18),
    ), // Text
  ],
), // Row
```



# 🚩 Flag 5 – Give It Some Space

**Objective:** Improve the spacing between your widgets.

Research:

`SizedBox()`

Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

**Challenge:** Make the card easier to read **without changing its colors or fonts yet**.

**Screenshot:**

- `SizedBox` usage
- Updated application

```
const Text(
  'Krylle Jane Y. Racaza',
  style: TextStyle(fontSize: 25,fontWeight: FontWeight.bold),
), // Text

const SizedBox(height: 10),

const Text(
  'BSIT - 3',
  style: TextStyle(fontSize: 20),
), // Text

const SizedBox(height: 10),

const Text(
  'My First Flutter Application',
  style: TextStyle(fontSize: 18),
  textAlign: TextAlign.center,
), // Text

const SizedBox(height: 10),

const Text(
  'August 11, 2026',
```

```
const Text(
  'August 11, 2026',
  style: TextStyle(fontSize: 18),
), // Text

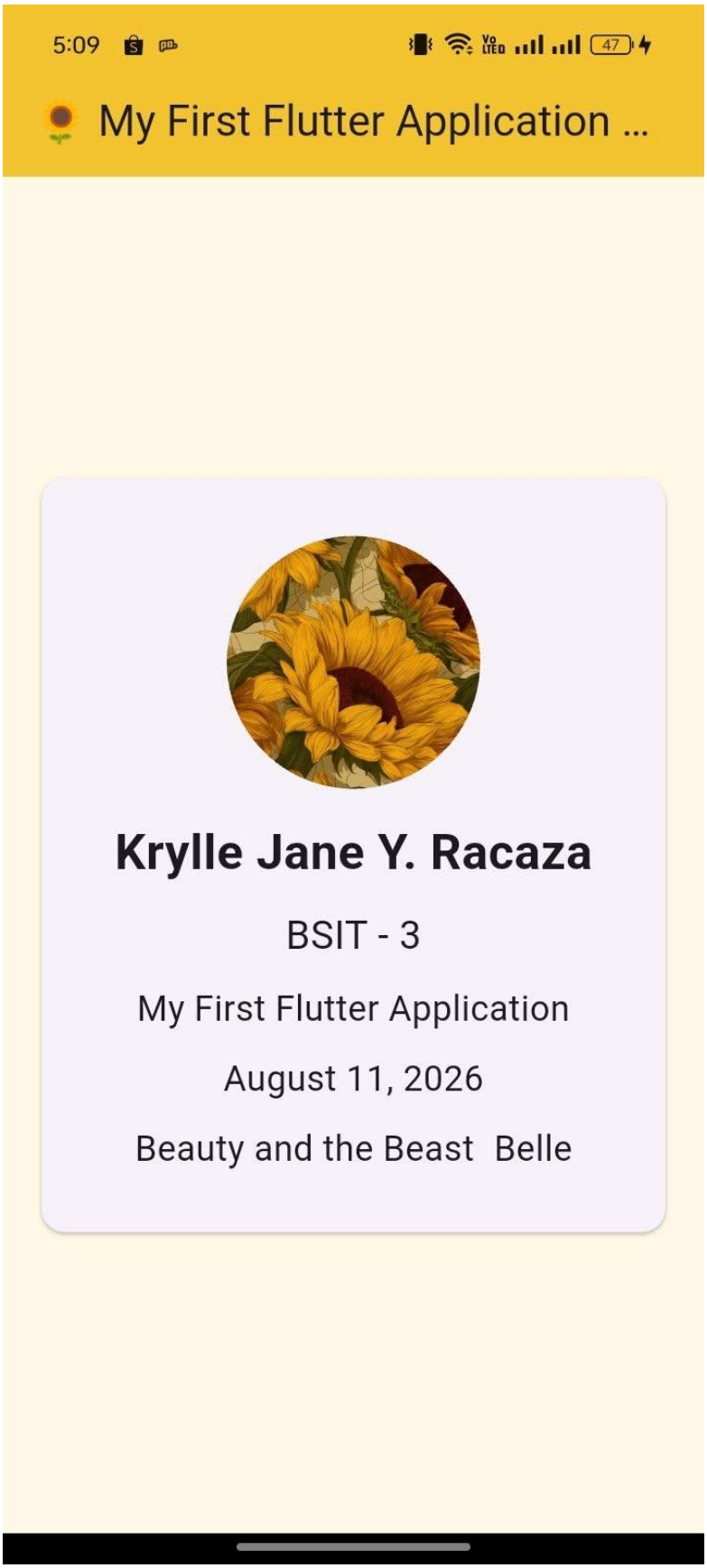
const SizedBox(height: 10),

Row(mainAxisAlignment: MainAxisAlignment.center,
  children: const [

    Text(
      'Beauty and the Beast',
      style: TextStyle(fontSize: 18),
    ), // Text

    SizedBox(width: 10),

    Text(
      'Belle',
      style: TextStyle(fontSize: 18),
    ), // Text
  ],
```



# 🚩 Flag 6 – Make It Look Better

**Objective:** Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

**Screenshot:**

- Your `TextStyle` changes
- Updated application

```
const Text(
  'My First Flutter Application',
  style: TextStyle(fontSize: 18, fontStyle: FontStyle.italic),
  textAlign: TextAlign.center,
), // Text

const SizedBox(height: 10),

const Text(
  'Krylle Jane Y. Racaza',
  style: TextStyle(fontSize: 25,fontWeight: FontWeight.bold),
), // Text

const SizedBox(height: 10),

const Text(
  'BSIT - 3',
  style: TextStyle(fontSize: 20),
), // Text

const SizedBox(height: 10),

const Text(
  'August 11, 2026',
  style: TextStyle(fontSize: 15),
), // Text

const SizedBox(height: 10),
```

5:18

My First Flutter Application ...



*My First Flutter Application*

**Krylle Jane Y. Racaza**

BSIT - 3

August 11, 2026

Beauty and the Beast  Belle

# 🚩 Flag 7 – Improve the Layout

**Objective:** Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

**Screenshot:**

- The properties you changed
- Updated application

```
child: Column(
  crossAxisAlignment: CrossAxisAlignment.stretch,
  children: [
```

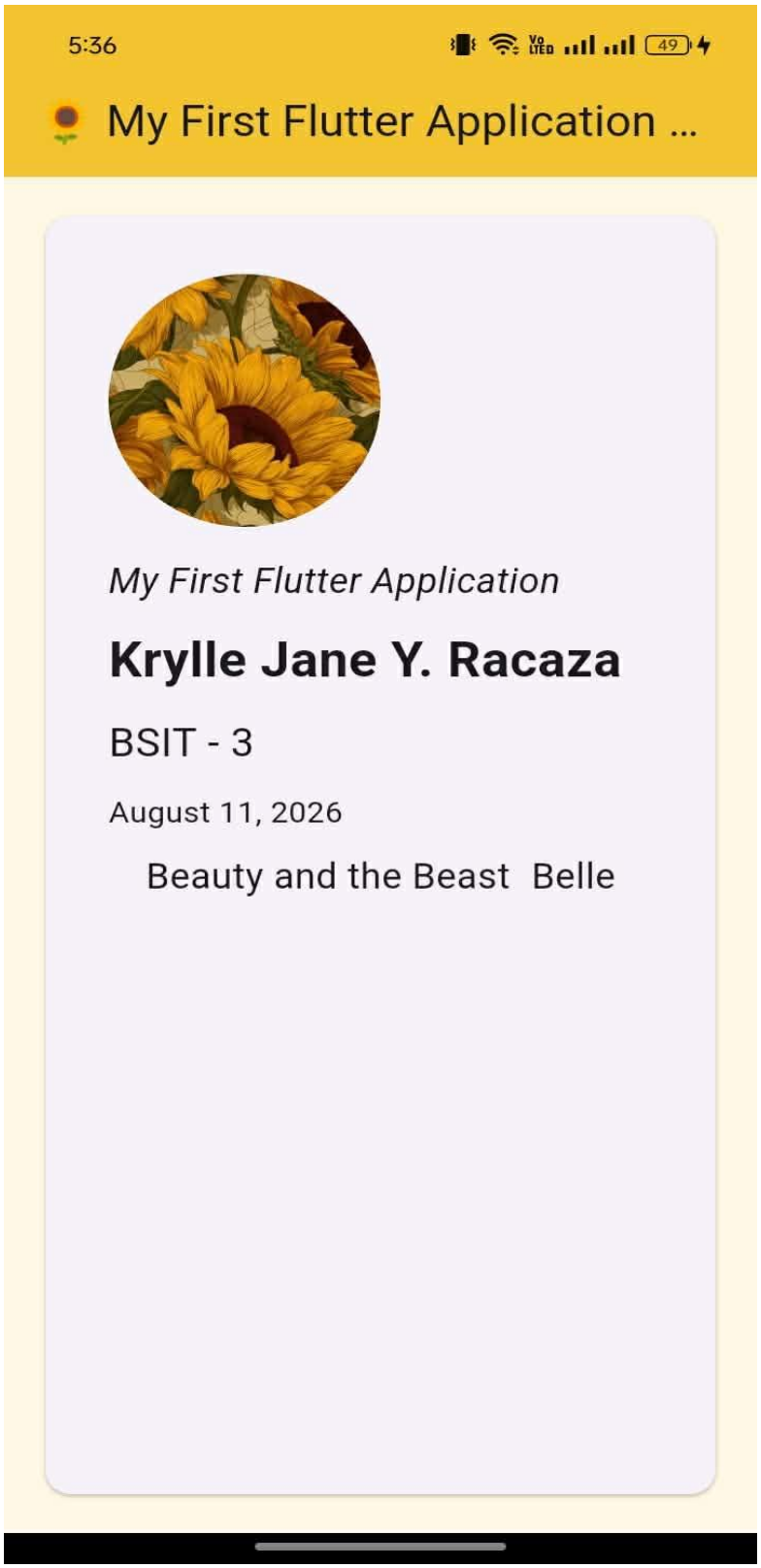
```
Row(mainAxisAlignment: MainAxisAlignment.center,
  children: const [

    Text(
      'Beauty and the Beast',
      style: TextStyle(fontSize: 18),
    ), // Text

    SizedBox(width: 10),

    Text(
      'Belle',
      style: TextStyle(fontSize: 18),
    ), // Text
```

Experiment:


5:36
My First Flutter Application ...
My First Flutter Application
Krylle Jane Y. Racaza
BSIT - 3
August 11, 2026
Beauty and the Beast Belle

# 🚩 Flag 8 – Create a Second Card

**Objective:** Apply what you learned to create another section.

Create a second `Card` containing something different from your profile.

For example:

## My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual `Text` widgets
- A `Column`
- At least one `Row`
- Spacing

**Screenshot:**

- Code for the second card
- Both cards displayed

```
body: Center(
  child: Card(
    margin: const EdgeInsets.all(20),
    child: Padding(
      padding: const EdgeInsets.all(30),

      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          Card(
    margin: const EdgeInsets.all(20),
    child: Padding(
      padding: const EdgeInsets.all(30),

      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

        const Text(
          'Wattpad',
          style: TextStyle(fontSize: 22),
        ), // Text
        const SizedBox(height: 10),

        const Text(
          'French Fries',
          style: TextStyle(fontSize: 22),
        ), // Text
        const SizedBox(height: 10),

        const Text(
          'Tangled',
          style: TextStyle(fontSize: 22),
        ), // Text
        const SizedBox(height: 10),
```

```
          const Text(
            'Tangled',
            style: TextStyle(fontSize: 22),
          ), // Text
          const SizedBox(height: 10),

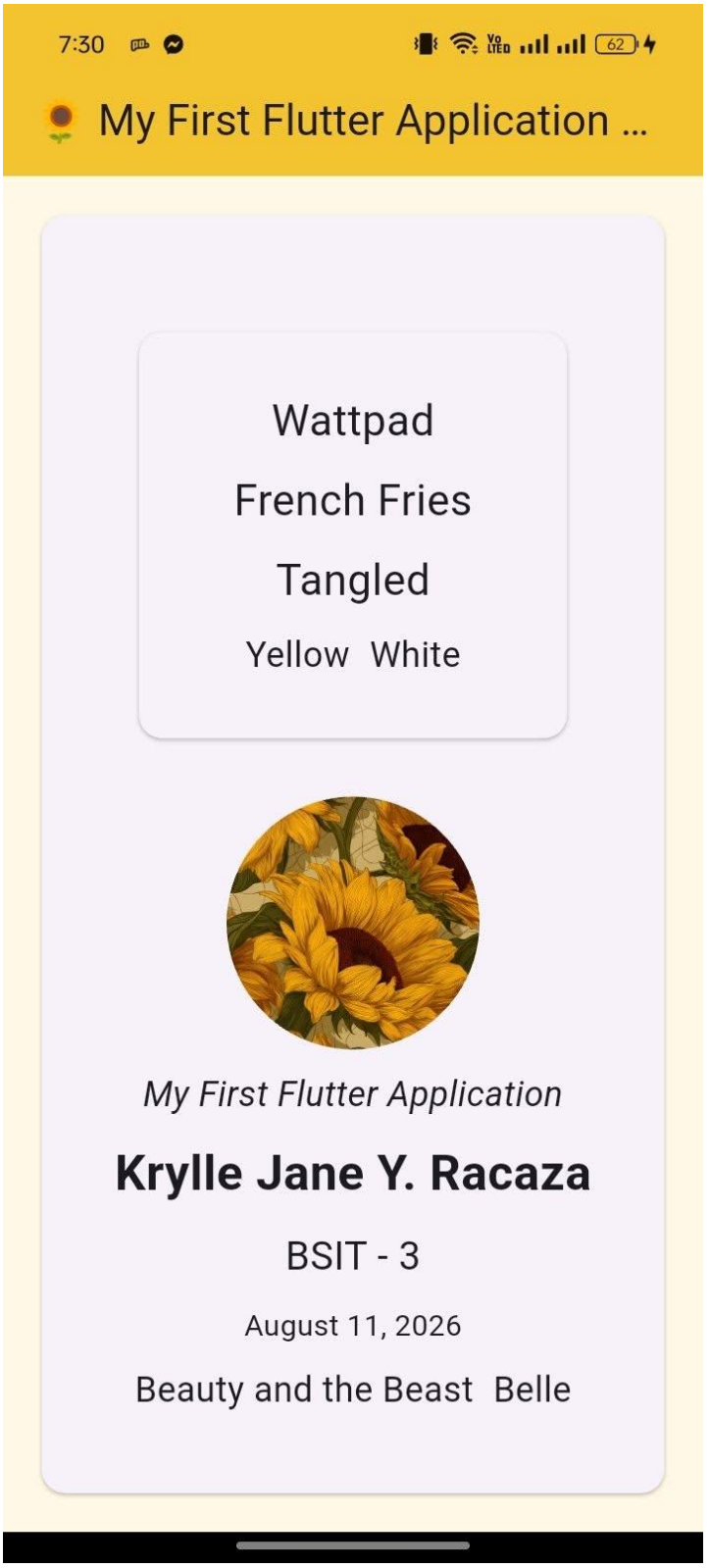
          Row(mainAxisAlignment: MainAxisAlignment.center,
            children: const [

              Text(
                'Yellow',
                style: TextStyle(fontSize: 18),
              ), // Text

              SizedBox(width: 10),

              Text(
                'White',
                style: TextStyle(fontSize: 18),
              ), // Text
            ],
          ), // Row
        ],
      ), // Column
    ), // Padding
  ), // Card

const SizedBox(height: 10),
```



# 🚩 Flag 9 – Stack the Cards

**Objective:** Arrange your two cards vertically.

You now have:

Profile Card

↓

Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

**Screenshot:**

- The widget containing both Cards
- Both cards visible in the application

```
    Card(
margin: const EdgeInsets.all(20),
child: Padding(
  padding: const EdgeInsets.all(30),

  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [

    const Text(
      'Wattpad',
      style: TextStyle(fontSize: 22),
    ), // Text
    const SizedBox(height: 10),

    const Text(
      'French Fries',
      style: TextStyle(fontSize: 22),
    ), // Text
    const SizedBox(height: 10),

    const Text(
      'Tangled',
      style: TextStyle(fontSize: 22),
    ), // Text
    const SizedBox(height: 10),

      Row(mainAxisAlignment: MainAxisAlignment.center,
        children: const [

        Text(
          'Yellow',
          style: TextStyle(fontSize: 18),
        ), // Text

        SizedBox(width: 10),
```

```
                    Text(
                      'White',
                      style: TextStyle(fontSize: 18),
                    ), // Text
                  ],
                ), // Row
              ],
            ), // Column
          ), // Padding
        ), // Card

        const SizedBox(height: 10),
                  ],
                ), // Column
              ), // Padding
            ), // Card
          ), // Center
        ), // Scaffold
      ); // MaterialApp
    }
}
```

My First Flutter Application ...



*My First Flutter Application*

**Krylle Jane Y. Racaza**

BSIT - 3

August 11, 2026

Beauty and the Beast  Belle

Wattpad

French Fries

Tangled

Yellow  White

# 🏴 Flag 10 – Widget Discovery

**Objective:** Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

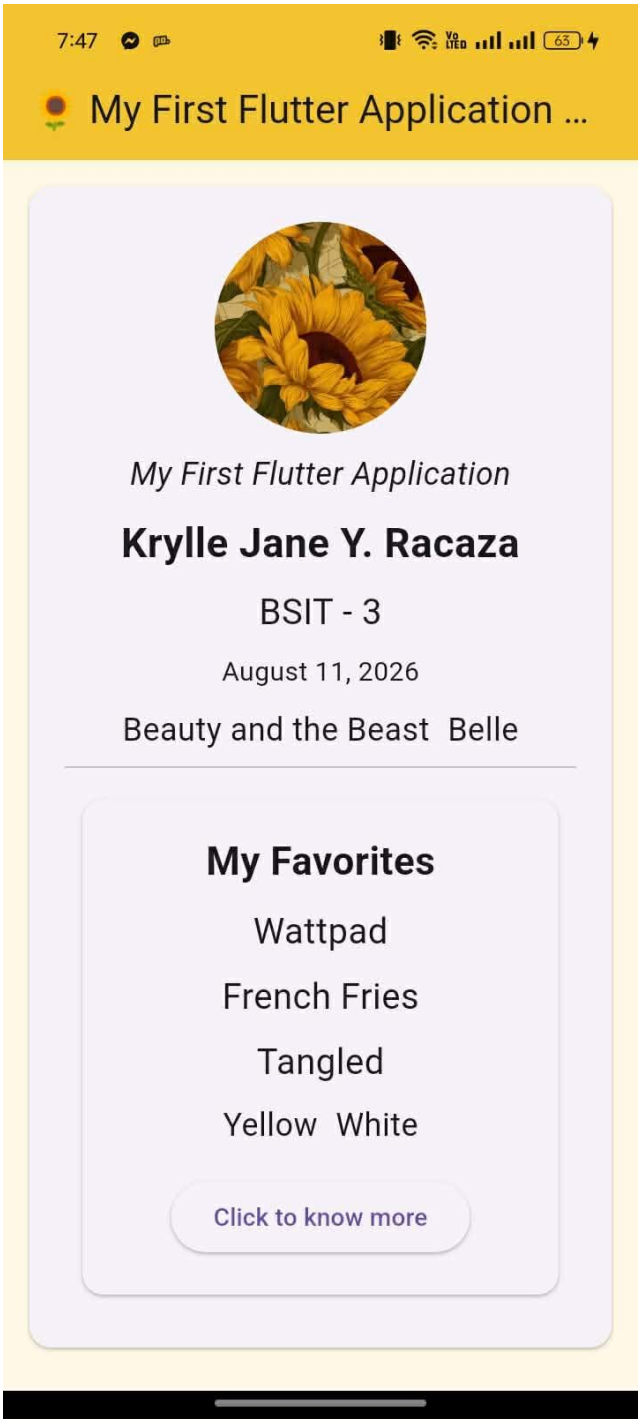
**Screenshot:**

- Your newly discovered widget in the code
- The widget visible in the application



```
const Divider(),
```

```
ElevatedButton(
  onPressed: () {},
  child: const Text(
    'Click to know more',
  ), // Text
), // ElevatedButton
```



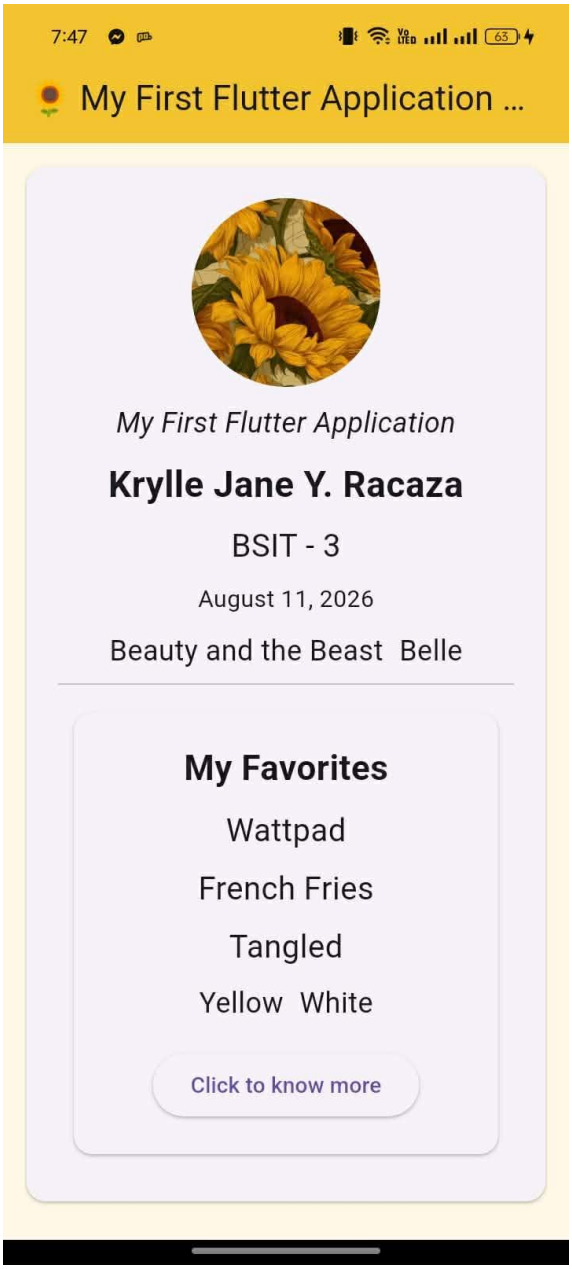
# 🏴 Flag 11 – Real Phone Test 📱

**Objective:** Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

**Screenshot:**

- Your completed application running on the physical phone (screenshot from your actual phone)



# 🏳️ Flag 12 – Save Your Progress

**Objective:** Commit your completed Widget branch.

Follow proper standards of commit messages

**Screenshot:**

- Successful commit
- `feature/widgets` branch

